# Markscheme

Mathematics: analysis and approaches
Practice question bank

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**161. (a)**

Consider the expression $\dfrac{(\cos 3\theta + i\sin 3\theta)(\cos 2\theta + i\sin 2\theta)^{-1}}{(\cos\theta + i\sin\theta)^4}$.

Using De Moivre's theorem, simplify this expression to the form $\cos(k\theta) + i\sin(k\theta)$, and state the value of $k$.

the numerator has argument $3\theta + (-2\theta) = \theta$ *(M1)* **A1**

dividing by $(\cos\theta + i\sin\theta)^4$ subtracts argument $4\theta$: $\theta - 4\theta$ *(M1)*

$$k = -3$$ **A1**

**[5 marks]**